Literature dissection

Spring 2026

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Photo documentation

Individually:

1. Find the document titled "Data Description: Photo Documentation of the North Campus Open Space Restoration Project".
2. Read about the photo documentation process.
3. Find 1-3 photo points relevant to your study using the ArcGIS web map.

As a group:

4. Discuss which photo point to focus on based on its relevance to your study question or context.
5. Decide on a photo point.
6. Find 2-4 photos at that photo point to represent seasonal changes within a water year.
7. In the table below, fill in the information for each photo at the photo point.

Table

Photo point number	Date of photo	Season	Direction of photo	Link to photo
18	April 2026	Spring	N	Photo 18 April 2026
18	October 2025	Fall	N	Photo 18 October 2025
18	April 2025	Spring	N	Photo 18 April 2025
20	April 2026	Spring	NW	Photo 20 April 2026
20	October 2025	Fall	NW	Photo 20 October 2025
20	April 2025	Spring	NW	Photo 20 April 2025
42	April 2026	Spring	S	Photo 42 April 2026
42	October 2025	Fall	S	Photo 42 October 2025
42	April 2025	Spring	S	Photo 42 April 2025

8. Describe in 5-7 sentences total:

- which water year you chose to examine
- how that water year is relevant to your study
- which photo point you chose to examine
- how that photo point is relevant to your study

- what changes you see in vegetation, animals, environmental characteristics from the photos you chose.

We chose to examine water year 2026 because it overlaps with the most recent vegetation monitoring period and helps us visually compare seasonal changes in salt marsh and brackish marsh vegetation at NCOS. This water year is relevant to our study because our project focuses on how salt marsh vegetation cover and plant community identity change across survey years, and the photo documentation gives us a visual reference for interpreting those patterns. We focused on photo points 18, 20, and 42 because together they show different views of the marsh plain, creek edge, riverbank, and surrounding salt marsh vegetation. Photo point 18 provides a broad view of the flat salt marsh vegetation, including sandy patches and a creek in the background, while photo points 20 and 42 show the north and south sides of the creek and help us compare vegetation conditions along the riverbank. Across the photos, vegetation appears much greener in April 2026 than in October 2025 at photo point 18, while October 2025 appears more gray-brown, likely reflecting drier late-season conditions. At photo points 20 and 42, April 2025 appears greener than April 2026, while October 2025 shows gray-brown vegetation and a larger area of accumulated sand along the riverbank. These seasonal and annual differences suggest that vegetation cover at the marsh edge can shift noticeably through time, likely reflecting changes in seasonal growth, water availability, sediment movement, and creek/riverbank conditions.

Reports or posters

Individually:

1. Navigate to a. the Posters section or b. the Reports section of the collection.
2. Find the poster(s) or report(s) related to your project.

As a group:

3. Discuss which poster or report to focus on based on its relevance to your study question or context. **Note that you only need one of either type.**
4. Read the poster or report.
5. Discuss the key insights.
6. Provide the information in the space below. For "Key insights" and "Relevance to your study", respond in 3-5 sentences.

Report or poster information

Title: North Campus Open Space Restoration Project Monitoring Report: Year 7, December 2024

Authors: ALison Rickard

Year published: 2024

Permalink: https://escholarship.org/uc/item/7bq618m8

Dataset used (likely): The report likely uses long-term NCOS restoration monitoring datasets collected by the Cheadle Center for Biodiversity and Ecological Restoration, including vegetation monitoring data similar to veg.csv and supporting vegetation transect metadata such as vp_veg_metadata.csv.

Key insights: The Year 7 monitoring report evaluates the progress of the North Campus Open Space restoration project across several habitat types, including wetland, upland, and transitional habitats (Rickard 2024). The report is useful for understanding how vegetation cover, native plant establishment, non-native cover, and broader habitat development are used as indicators of restoration progress at NCOS. It also provides the larger restoration context for our project, showing that vegetation monitoring is not only a way to describe plant cover, but also a way to track whether restored habitats are developing toward their intended ecological conditions.

Relevance to your study (be specific here with the question you have asked, the figures you've made, the insights you've gained, etc.): This report is directly relevant to our project because our study focuses on salt marsh and closely related brackish marsh vegetation at NCOS (Rickard 2024). Our figures build on the same general monitoring goals by asking which plant species characterize the marsh, how dominant species change through time, and how native and non-native vegetation contribute to habitat identity. While the monitoring report summarizes restoration progress more broadly across NCOS, our project narrows in on species-level vegetation patterns within the salt marsh/brackish marsh habitat and uses the vegetation dataset to explore those patterns more closely.

Annotated bibliography

Background

You can find peer-reviewed articles by searching on Google Scholar, which is a large database of research papers, posters, and other materials.

Note: while it may be tempting to use AI to provide a list of papers for you, you should *not* be using AI for any component of this activity. You'll accomplish the objective of articulating the connection between your study and other study systems/questions by completing the difficult task of sifting through papers that may not be relevant to you before finding one that is.

Additionally, you can test out inserting citations in Quarto. See this resource to insert a citation in a .qmd.

Directions

Individually:

1. Search Google Scholar for papers that might be relevant to your study. One recommendation for how to start: search for whatever your focal variables may be + "wetland restoration" (for example, "bird abundance wetland restoration", "water salinity wetland restoration", "aquatic invertebrate wetland restoration").
2. Find 4-5 papers that might be relevant based on title and what shows up in the search.
3. Read the abstracts of those 4-5 papers.
4. Decide which - if any - are actually relevant to your project. Make a note of these papers. Note that a paper may be relevant in that it shares a) taxonomic groups, b) geographic regions, c) research questions, d) research methodology - there are lots of ways that existing research can be tied into your project.
5. If step 4 did not yield any relevant papers, repeat steps 2-4.
6. Search for papers until you have 4 relevant papers per person (12 total for a 3 person group).

As a group:

7. Share the papers you found, discussing the main points of the paper and how they are relevant to your project.
8. Fill in the template for an annotated bibliography below.
9. In the "Main findings" or "Relevance to study" sections, insert a citation for the paper.

Paper 1

Title: Restoration of urban salt marshes: Lessons from southern California

Authors: John C. Callaway and Joy B. Zedler

Main findings (2-4 sentences): Callaway and Zedler describe several major challenges for restoring urban salt marshes in southern California, including habitat isolation, fragmentation, exotic species, loss of transitional upland habitat, and altered hydrologic and sediment dynamics (Callaway and Zedler 2004). They emphasize that urban salt marshes are strongly shaped by surrounding land use, watershed conditions, and the ability of native plants to colonize restored sites. The paper also argues that low species richness can slow the development of important marsh functions such as biomass accumulation and nitrogen retention.

Relevance to study (2-4 sentences): This paper is highly relevant because NCOS is a restored coastal wetland embedded within an urban and university landscape. Our project asks which species characterize salt marsh habitat at NCOS and how native and non-native vegetation contribute to marsh identity. Callaway and Zedler help frame why species composition matters in restored urban salt marshes, especially when exotic species, hydrologic modification, and habitat fragmentation can influence restoration trajectories (Callaway and Zedler 2004).

Paper 2

Title: Large-scale field studies inform adaptive management of California wetland restoration

Authors: Kathryn M. Beheshti, Stephen C. Schroeter, Andres A. Deza, Daniel C. Reed, Rachel S. Smith, and Henry M. Page

Main findings (2-4 sentences): Beheshti et al. studied vegetation establishment in a restored southern California salt marsh and found that plant cover was strongly influenced by physical stressors such as soil salinity, soil moisture, elevation, and soil compaction (Beheshti et al. 2023). Their field experiments showed that restoration outcomes can vary by marsh zone and by plant species, with Salicornia pacifica naturally recruiting and increasing cover in some areas. The study highlights how adaptive management and field experiments can help identify the factors that limit vegetation establishment in restored wetlands.

Relevance to study (2-4 sentences): This paper supports our use of percent cover as an important restoration indicator because it shows that vegetation establishment is central to wetland restoration success. It also helps us interpret changes in dominant plant cover at NCOS as potential signals of restoration trajectory rather than just visual patterns. Since our project examines mean percent cover through time, Beheshti et al. gives us a strong scientific basis for connecting vegetation cover to soil conditions, hydrology, and restoration management (Beheshti et al. 2023).

Paper 3

Title: Relationship between topographic heterogeneity and vegetation patterns in a Californian salt marsh

Authors: H. Morzaria-Luna, J. C. Callaway, G. Sullivan, and J. B. Zedler

Main findings (2-4 sentences): Morzaria-Luna et al. examined how salt marsh vegetation richness, cover, and assemblage composition differed between areas with and without tidal creeks (Morzaria-Luna et al. 2004). They found that species richness, frequency, cover, and assemblages varied with topographic heterogeneity, especially the presence of tidal creeks. The authors concluded that restoration of Californian salt marshes should target multi-species assemblages rather than monocultures.

Relevance to study (2-4 sentences): This paper connects directly to our question about which species characterize salt marsh habitat at NCOS. Our project uses species ranking and dominant-species visualizations to describe the plant assemblage of the restored marsh, and this paper supports the idea that marsh identity is shaped by multiple species rather than a single dominant plant. It also helps explain why creek edges, marsh plains, and topographic variation may influence the vegetation patterns we see in our figures (Morzaria-Luna et al. 2004).

Paper 4

Title: Causes and consequences of invasive plants in wetlands: Opportunities, opportunists, and outcomes

Authors: Joy B. Zedler and Suzanne Kercher

Main findings (2-4 sentences): Zedler and Kercher argue that wetlands are especially vulnerable to plant invasions because they often act as landscape sinks, accumulating water, nutrients, sediment, debris, and other materials that can create opportunities for opportunistic species (Zedler and Kercher 2004). They explain that invasive wetland plants can form monotypes, alter habitat structure, reduce biodiversity, change nutrient cycling, and modify food webs. The paper also emphasizes that invasion is often linked to disturbance, propagule pressure, hydrologic alteration, and resource availability.

Relevance to study (2-4 sentences): This paper is important for our native/non-native and dominant non-native species figures. It helps explain why non-native cover matters ecologically and why we should look beyond total vegetation cover to ask which species are contributing to habitat structure. In our project, tracking non-native species through time helps us interpret whether restored salt marsh habitat is becoming more strongly characterized by native halophytes or whether non-native plants are contributing substantially to vegetation cover (Zedler and Kercher 2004).

Paper 5

Title: Carpinteria Salt Marsh: Environment, History, and Botanical Resources of a Southern California Estuary

Authors: Wayne R. Ferren, Jr.

Main findings (2-4 sentences): Ferren provides a detailed botanical and environmental account of Carpinteria Salt Marsh, including its physical setting, history, wetland vegetation types, plant inventory, and disturbance history (Ferren 1985). The report describes southern California estuaries as dynamic systems shaped by salinity, freshwater runoff, seasonal rainfall, tidal circulation, sedimentation, and historical land use. It also documents many characteristic salt marsh plants and wetland vegetation communities in the Santa Barbara region.

Relevance to study (2-4 sentences): This source is valuable because it gives our project a local botanical reference point from a nearby Santa Barbara County salt marsh. Although our study focuses on NCOS, Ferren helps us understand the broader regional context for southern California salt marsh vegetation and the environmental pressures that shape these communities. It is especially useful for interpreting characteristic marsh plants such as pickleweed, saltgrass, saltbushes, and other halophytes within a local coastal wetland framework (Ferren 1985).

Paper 6

Title: Monitoring vegetation dynamics at a tidal marsh restoration site: Integrating field methods, remote sensing and modeling

Authors: Alexandra S. Thomsen, Johannes Krause, Monica Appiano, Karen E. Tanner, Charlie Endris, John Haskins, Elizabeth Watson, Andrea Woolfolk, Monique C. Fountain, and Kerstin Wasson

Main findings (2-4 sentences): Thomsen et al. monitored vegetation colonization at a restored tidal marsh using field surveys, UAS imagery, and modeling (Thomsen et al. 2021). They found that vegetation establishment was strongly influenced by elevation, inundation frequency, salinity stress, and soil conditions. The paper also highlights that vegetation cover is commonly used as a restoration monitoring metric because it reflects the establishment of foundation species that support broader marsh functions.

Relevance to study (2-4 sentences): This paper is relevant because our project also uses vegetation cover to evaluate restoration patterns through time. Although we are working with field-based vegetation monitoring data rather than UAS imagery, Thomsen et al. supports the idea that vegetation monitoring can reveal where restoration is progressing and where environmental conditions may limit plant colonization. It also connects well to our descriptive figures, which use percent cover to compare dominant species, native and non-native vegetation, and total vegetation cover across survey years (Thomsen et al. 2021).

Paper 7

Title: UC Santa Barbara's Role in Protecting the Upper Devereux Slough

Authors: Duncan A. Mellichamp

Main findings (2-4 sentences): Mellichamp provides a historical account of the planning, acquisition, and restoration of the North Campus Open Space, tracing the project from the filling of the upper Devereux Slough to construct the Ocean Meadows Golf Course in the 1960s through its eventual transfer to UCSB and restoration by the Cheadle Center (Mellichamp 2023). The report describes a multi-decade collaborative planning effort among UCSB, Santa Barbara County, the City of Goleta, and The Trust for Public Land, which ultimately secured approximately \$18 million in public funding for wetland reconstruction and habitat restoration. Mellichamp emphasizes that the restoration was intended to undo the ecological damage caused by converting the wetland to a golf course, returning the site to a complex of habitats including wetlands, uplands, and transitional zones that support special-status plants and animals.

Relevance to study (2-4 sentences): This source provides the foundational historical and institutional context for our project by documenting why and how the salt marsh habitat we are analyzing came to exist at NCOS (Mellichamp 2023). Understanding that the site was actively restored from a filled wetland helps us interpret the vegetation patterns in our figures, especially the dominant native halophytes we identify, such as Salicornia pacifica and Distichlis spicata, represent the return of coastal salt marsh community structure to a landscape that was essentially removed in the 1960s.

Paper 8

Title: A simple empirical model of salt marsh plant spatial distributions with respect to a tidal channel network

Authors: Eric W. Sanderson, Theodore C. Foin, and Susan L. Ustin

Main findings (2-4 sentences): Sanderson et al. developed a GIS-based model using a cumulative inverse squared distance function to quantify tidal channel influence at each point in a California salt marsh, finding that this single spatial factor could predict the distributions of both dominant and minor plant species across the marsh (Sanderson et al. 2001). The model showed that Salicornia dominated areas far from large channels, while species such as Distichlis, Frankenia, and Jaumea were associated with intermediate-sized channels, and Spartina occupied the zone of highest channel influence closest to the river. These results emphasize that tidal channel networks are a primary structural force shaping salt marsh plant community composition and spatial zonation.

Relevance to study (2-4 sentences): This paper is relevant to our project because it demonstrates that the same species we identify as dominant at NCOS are characteristically zoned with respect to tidal hydrology in California salt marshes (Sanderson et al. 2001). While our project does not model spatial distributions, Sanderson et al. shows why these species appear where they do, which helps us interpret shifts in dominant species cover at NCOS as potential signals of changing hydrologic conditions or restoration trajectory. The study also supports our focus on species identity rather than total cover alone, since different species occupy distinct positions along the tidal gradient and contribute differently to marsh structure.

Paper 9

Title: Status of Ventura marsh milk-vetch at North Campus Open Space

Authors: Lisa Stratton, Wayne Chapman, Carlo Calderon

Main findings (2-4 sentences): Stratton et al. report on the outplanting and establishment of Ventura marsh milk-vetch (Astragalus pycnostachyus var. lanosissimus), a federally endangered species, at NCOS, finding that the majority of outplanted seedlings thrived and successfully set seed under ambient conditions when placed on sandy ridges and slopes rather than in swales subject to flooding (Stratton et al. 2021). Plant performance was strongly tied to microtopography and hydrology: seedlings on ridges and slopes showed significantly greater height, vigor, and seed production compared to those in swales, where 55 percent died following high estuary water levels and sediment inundation. The report also documents successful natural recruitment of more than 2,000 seedlings adjacent to the primary outplanting site by spring 2021, indicating that NCOS can support self-sustaining populations of this rare species under appropriate hydrologic and soil conditions.

Relevance to study (2-4 sentences): This report is relevant because it demonstrates how strongly hydrology, soil moisture, and microtopography shape plant establishment and survival at NCOS (the same site where our vegetation monitoring data were collected) (Stratton et al. 2021). While our project focuses on salt marsh vegetation cover and species composition rather than a single rare species, Stratton et al. reinforce that the patterns we observe in our figures are not just statistical summaries but reflect real ecological processes tied to water availability, flooding regime, and site conditions. The report also adds important context about NCOS as an active restoration site where plant community development is shaped by ongoing management decisions, hydrologic dynamics, and species-specific responses to environmental conditions.

Paper 10

Title: Ventura Marsh Milk-vetch (Astragalus pycnostachyus var. lanosissimus) 2024 Management and Monitoring Report

Authors: Claire Wilhelm-Safian, Wayne Chapman, and Lisa Stratton

Main findings (2-4 sentences): Wilhelm-Safian et al. report that the Ventura marsh milk-vetch population at NCOS declined substantially in 2024, with adult counts dropping from an estimated 814 to 308 individuals in the main population and total seedling abundance falling sharply across nearly all site zones (Wilhelm-Safian et al. 2024). The report identifies non-native nitrogen-fixing plants, particularly Melilotus indicus and Polypogon monspeliensis, as a growing threat, finding that nitrogen accumulated from milk-vetch's own nitrogen-fixation may be creating a positive feedback loop that benefits invasive weed competitors and degrades the low-nutrient sandy habitat that the milk-vetch requires. Population decline was also linked to two successive wet winters, increased flooding, reduced active management, and natural ecological succession, as bare sandy areas colonized by vegetation progressively reduced available habitat for the species.

Relevance to study (2-4 sentences): This report is relevant because it directly documents how non-native plant cover is expanding at NCOS in response to changing soil and hydrologic conditions, providing a site-specific example of the ecological consequences of non-native vegetation accumulation that our project addresses through broader vegetation monitoring (Wilhelm-Safian et al. 2024). The report reinforces our focus on tracking non-native species through time. It also contextualizes the NCOS landscape as a dynamic, actively managed site where vegetation patterns are shaped by ongoing disturbance, succession, management interventions, and interannual rainfall variability.

Paper 11

Title: The influence of rainfall variability on the species composition of a northern California salt marsh plant assemblage

Authors: Susan K. Allison

Main findings (2-4 sentences): Allison examined how interannual variation in rainfall affected plant species composition in a northern California salt marsh over four years, finding that in drier years Salicornia virginica increased its dominance while higher-rainfall years allowed less salt-tolerant species to expand their cover (Allison 1992). The study showed that species composition was relatively stable across the monitoring period despite year-to-year climatic variability, but that dominant species shifted in relative abundance in response to rainfall and associated changes in soil salinity and moisture. These findings suggest that even modest interannual climate variation can create detectable fluctuations in which species contribute most to marsh vegetation cover.

Relevance to study (2-4 sentences): This paper is directly relevant to our project because our figures track mean percent cover of dominant species across survey years spanning variable rainfall conditions in southern California (Allison 1992). Allison's findings help us interpret interannual variation in our dominant-species figure not solely as a restoration signal but also as a potential climate-driven response, particularly for Salicornia pacifica, which is expected to increase cover in drier years. The paper also supports our rationale for averaging across transects and years rather than treating any single year's vegetation pattern as definitive.

Paper 12

Title: Dominance of native grasses leads to community convergence in wetland restoration

Authors: Laurel Pfeifer-Meister, Bitty A. Roy, Bart R. Johnson, Jeff Krueger & Scott D. Bridgham

Main findings (2-4 sentences): This study conducted a three-year replicated field experiment in Oregon wetland prairie restoration, finding that plant community composition converged across different site preparation treatments over time as annual exotic species declined and native perennial grasses increased in dominance (**Pfeifer-Meister_2012?**). Community composition became progressively more similar to reference wetlands each year, but species richness and diversity declined as dominant natives outcompeted other species, revealing a trade-off between native cover and biodiversity. The authors argue that managing successional trajectories actively is necessary to achieve both high native dominance and diversity in restored wetlands.

Relevance to study (2-4 sentences): This paper is relevant because our project tracks how dominant species cover changes across survey years at NCOS, and Gonzalves et al. provide a broader conceptual framework for interpreting that trajectory, specifically, that increasing native dominance over time is a predictable and meaningful signal of restoration progress (**Pfeifer-Meister_2012?**). It also helps contextualize why a few dominant species like Salicornia pacifica may contribute a large and growing share of total vegetation cover without that being cause for concern. The study reinforces that tracking which species are dominant, and whether those species are native, is a more informative restoration metric than total cover alone.

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